

1. List Operations

Create a list of 5 fruits. Perform the following operations:

- Add a new fruit
- Remove one fruit
- Replace the second fruit with another
- Print the final list and its length

2. Tuple Analysis

Create a tuple with some duplicate values.

- Print the tuple
- Count how many times a specific value appears
- Find the index of a specific value
- Try modifying an element and observe the error

3. Set Behavior

Create a set with some duplicate values.

- Print the set
- Add a new element
- Remove an element using both `remove()` and `discard()`
- Try adding a list to the set and explain the error

4. Dictionary Manipulation

Create a dictionary with keys: name, email, and phone.

- Print all keys and values
- Update the name
- Add a new key address
- Access a value using `get()`

5. Nested Dictionary

Create a nested dictionary representing a student with subjects and scores.

- Access a nested value
- Add a new subject
- Print all keys and values

6. Unpacking and * Operator

Create a list of 5 vegetables and unpack the first two into variables and the rest into another list using `*`.

- Print all unpacked variables

7. Membership and Identity Operators

- Create a list and check if a value exists using `in`
- Create two identical lists and compare them using `is` and `==`
- Explain the difference between `is` and `==` in your own words

8. Shallow vs Deep Copy

- Create a nested list
- Make a shallow copy and modify an inner element
- Make a deep copy and modify an inner element
- Print all lists and explain the difference

9. Using Logical and Bitwise Operators

- Use `and`, `or`, and `not` with some conditions
- Use `&`, `|`, `^`, `~`, `<<`, and `>>` with integers
- Explain the output of each operation

10. Looping Through Data Structures

- Loop through a list and print each item
- Loop through a dictionary and print keys, values, and items
- Loop through a list of tuples and unpack each tuple